



Data Glacier

Your Deep Learning Partner

Exploratory Data Analysis

PROJECT : Healthcare - Persistency of a drug (Data Science)

BATCH CODE : LISUM49

Prepared by: Magdalena Ivanova

Group Name: DataVision

University: IMC Krems

Email: magdalena.ivanova.ds@gmail.com

GitHub repo link: https://github.com/magdalena-ivanova-ds/Data-Glacier-Virtual-Internship/tree/main/week13_Final_Project

Date: 24 November 2025

Agenda

Executive Summary

Overview of the Healthcare – Drug Persistency Analysis project

Problem Statement

Identifying factors influencing patient medication persistency

Approach

Data preparation, exploratory analysis, and model development

EDA

Analysis of demographics, physician, and adherence variables

EDA Summary

Key findings and patterns impacting persistency outcomes

Recommendations

Strategic actions to improve patient adherence and treatment consistency

Background & Purpose

The pharmaceutical industry faces a major challenge - ensuring patients continue taking their prescribed medications as instructed.

When patients discontinue early, it leads to:

- Poorer health outcomes
- Increased healthcare costs
- Treatment inefficiencies

ABC Pharma initiated this project to identify the drivers of drug persistency using a data-driven approach.

By understanding which patient, physician, and risk factors influence medication adherence, the company can improve patient care and reduce losses.



Project Objective



Goal: Develop a classification model that predicts whether a patient will remain persistent (1) or non-persistent (0) with prescribed medication.

The analysis focuses on:

- **Patient Demographics:** Age, Gender, Race, Ethnicity, Region
- **Physician Details:** Specialty, Specialist Flag, Category
- **Risk and Adherence Factors:** Risk segment, T-score change, Adherence flag

The outcome enables early identification of patients likely to stop treatment, supporting targeted interventions.

Dataset Overview

The dataset *Healthcare_dataset.xlsx* includes:

- 3 424 patient records
- 27 attributes describing demographics, physicians, and treatment factors
- No missing values - all entries complete
- Each record represents a patient's full profile, combining medical and behavioral information.

The clean and complete dataset allowed for detailed analysis without major data correction.

```
data.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3424 entries, 0 to 3423
Data columns (total 69 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0   Ptid                                       3424 non-null   object
1   Persistency_Flag                         3424 non-null   object
2   Gender                                   3424 non-null   object
3   Race                                     3424 non-null   object
4   Ethnicity                               3424 non-null   object
5   Region                                   3424 non-null   object
6   Age_Bucket                              3424 non-null   object
7   Ntm_Speciality                          3424 non-null   object
8   Ntm_Specialist_Flag                     3424 non-null   object
9   Ntm_Speciality_Bucket                   3424 non-null   object
10  Gluco_Record_Prior_Ntm                  3424 non-null   int64
11  Gluco_Record_During_Rx                  3424 non-null   int64
12  Dexa_Freq_During_Rx                     3424 non-null   int64
13  Dexa_During_Rx                          3424 non-null   int64
14  Frag_Frac_Prior_Ntm                     3424 non-null   int64
15  Frag_Frac_During_Rx                     3424 non-null   int64
16  Risk_Segment_Prior_Ntm                   3424 non-null   object
17  Tscore_Bucket_Prior_Ntm                  3424 non-null   object
18  Risk_Segment_During_Rx                   3424 non-null   object
19  Tscore_Bucket_During_Rx                  3424 non-null   object
...
67  Risk_Recurring_Falls                     3424 non-null   int64
68  Count_Of_Risks                           3424 non-null   int64
dtypes: int64(52), object(17)
```

Data Preparation

To prepare the dataset for analysis and modeling:

- All categorical fields labeled “Y” or “N” were converted to 1/0 values.
- Data types were standardized across all features.
- Duplicate checks confirmed dataset uniqueness.
- Variables were structured for compatibility with visualization and machine learning.

Outcome:

A clean, validated dataset ready for exploratory analysis and predictive modeling.

```
# Check if there are any duplicate values:

duplicates = data.duplicated()

if duplicates.any():
    print("Duplicates exist in the DataFrame.")
else:
    print("No duplicates found in the DataFrame.")

143] .. No duplicates found in the DataFrame.

# Double check if there are any null (NA) values:

Null_Values = data.isnull().sum()

if Null_Values.any():
    print("Null values exist in the DataFrame.")
else:
    print("No null values exist in the DataFrame.")

144] .. No null values exist in the DataFrame.
```

Data Preparation

```
data = pd.read_csv('Healthcare.csv')
data.head(5)
```

t	...	Risk_Family_History_Of_Osteoporosis	Risk_Low_Calcium_Intake	Risk_Vitamin_D_Insufficiency	Risk_Poor_Health_Frailty	Risk_Excessive_Thinness
1	...	N	N	N	N	N
1	...	N	N	N	N	N
1	...	N	Y	N	N	N
1	...	N	N	N	N	N
1	...	N	N	N	N	N

```
data = data.replace('N', 0)
data = data.replace('Y', 1)
data.head()
```

...	Risk_Family_History_Of_Osteoporosis	Risk_Low_Calcium_Intake	Risk_Vitamin_D_Insufficiency	Risk_Poor_Health_Frailty	Risk_Excessive_Thinness
...	0	0	0	0	0
...	0	0	0	0	0
...	0	1	0	0	0
...	0	0	0	0	0
...	0	0	0	0	0



Analysis Approach

My exploratory data analysis (EDA) examined relationships and patterns across:

- Numerical Values
- Patient Demographics
- Physician/Provider Attributes
- Risk and Adherence Factors

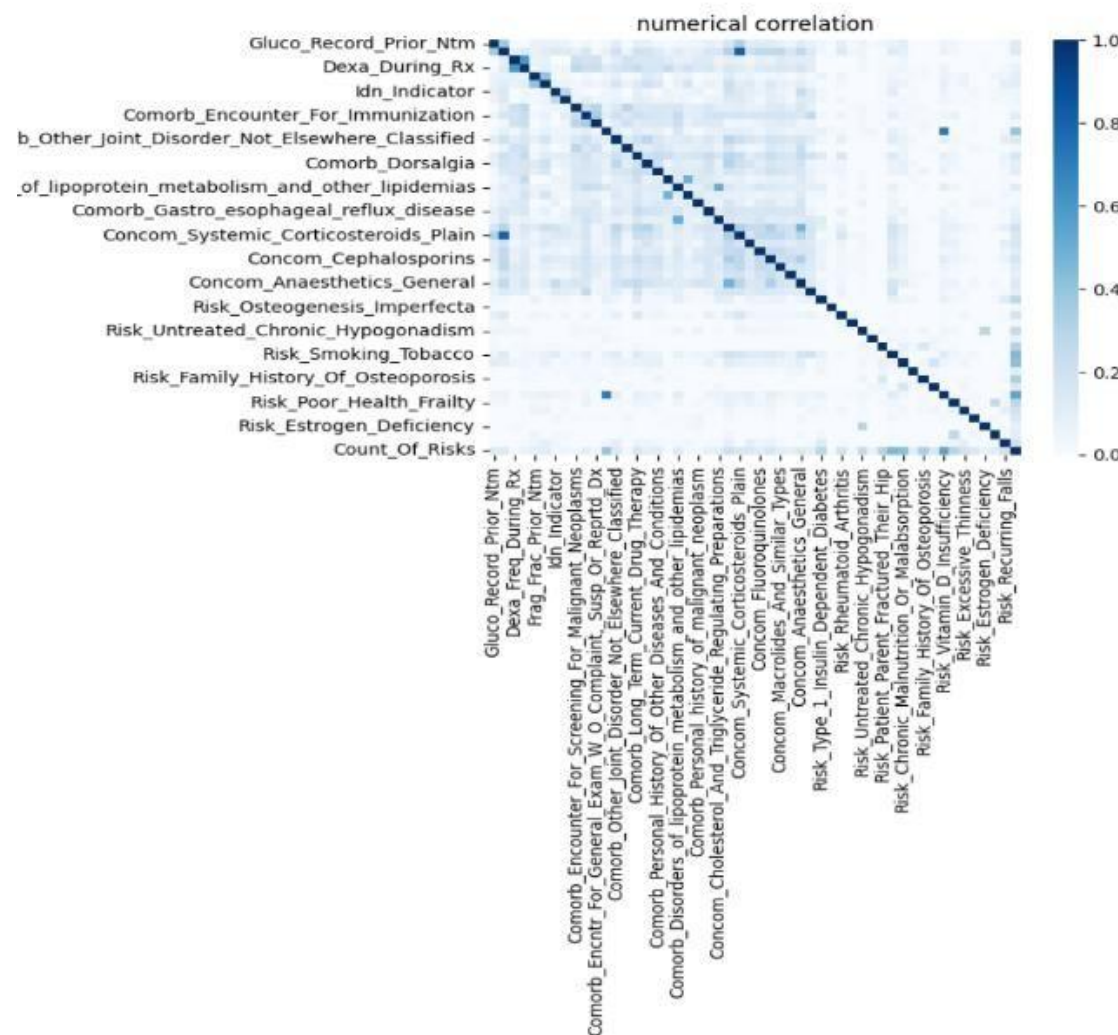
Each layer revealed behavioral and clinical characteristics affecting whether a patient remains consistent with treatment.

Numerical Insights

- Correlation among numeric variables was weak - indicating that purely numerical features (like age or T-score) do not strongly determine persistency.
- This finding shifted the focus toward categorical factors such as gender, region, and provider specialty, which appear more predictive of patient behavior.

Implication:

Human-related and contextual attributes drive adherence more than lab values alone.



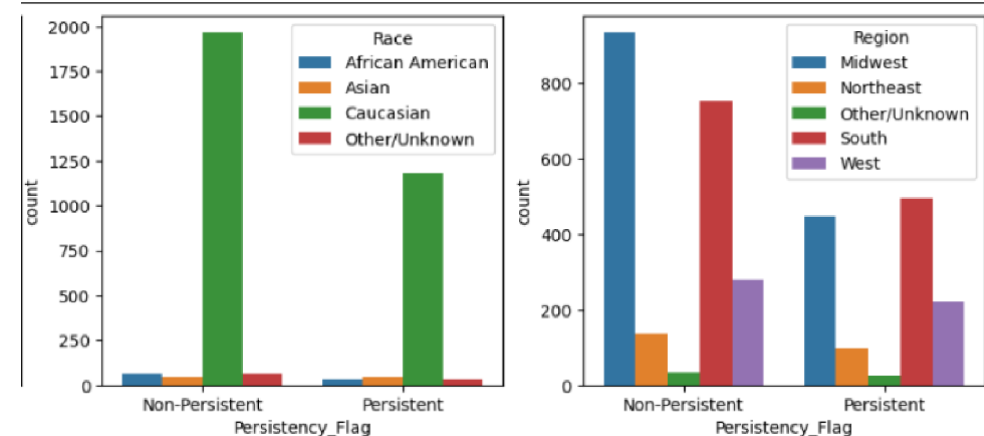
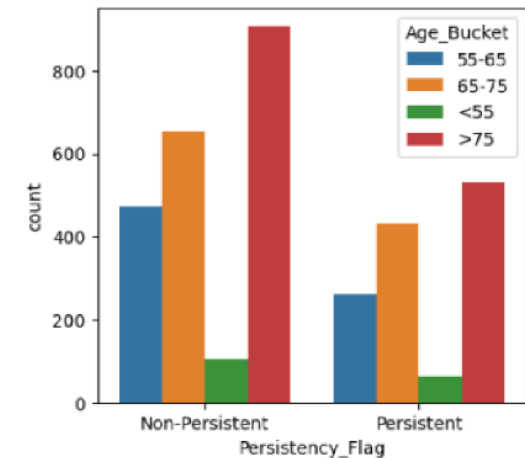
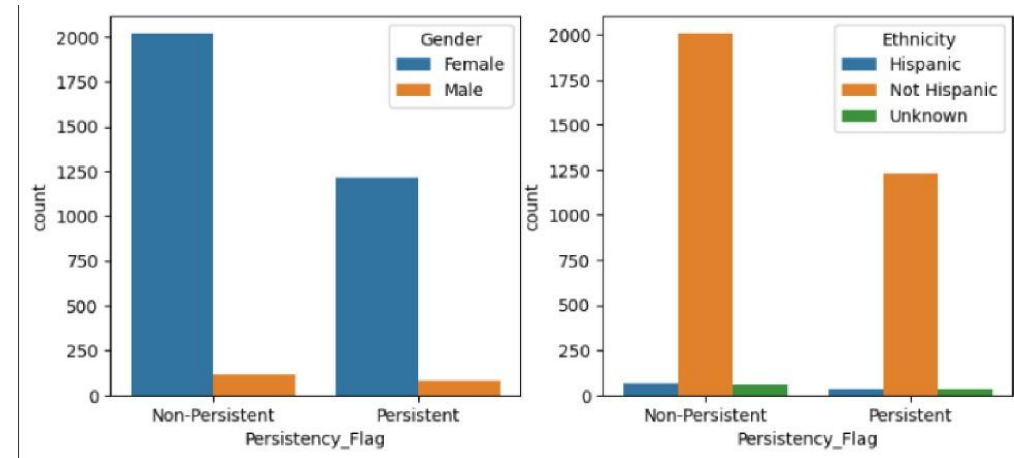
Patient Demographics

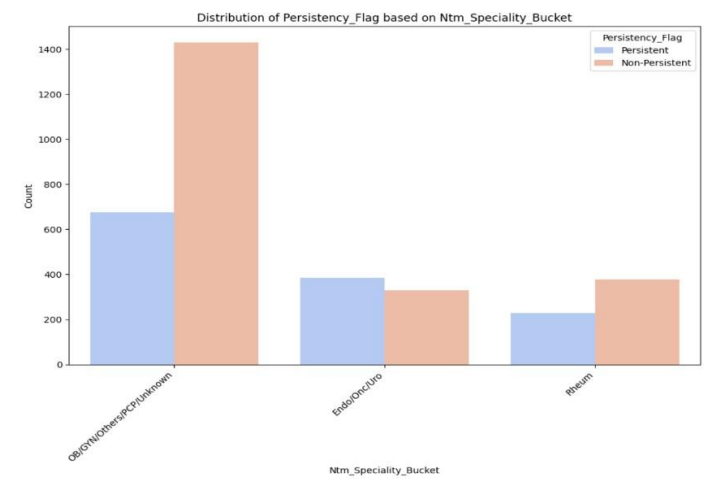
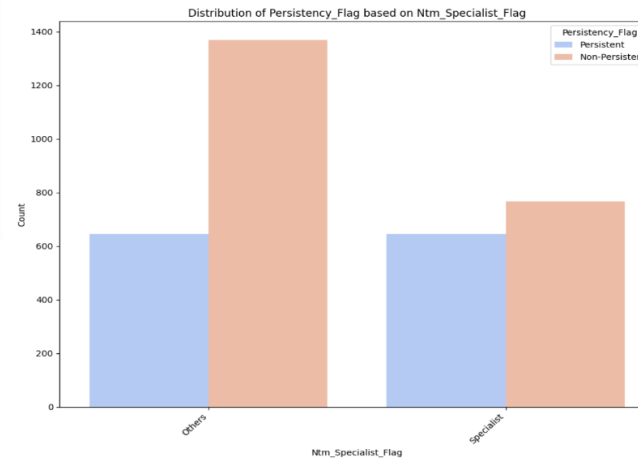
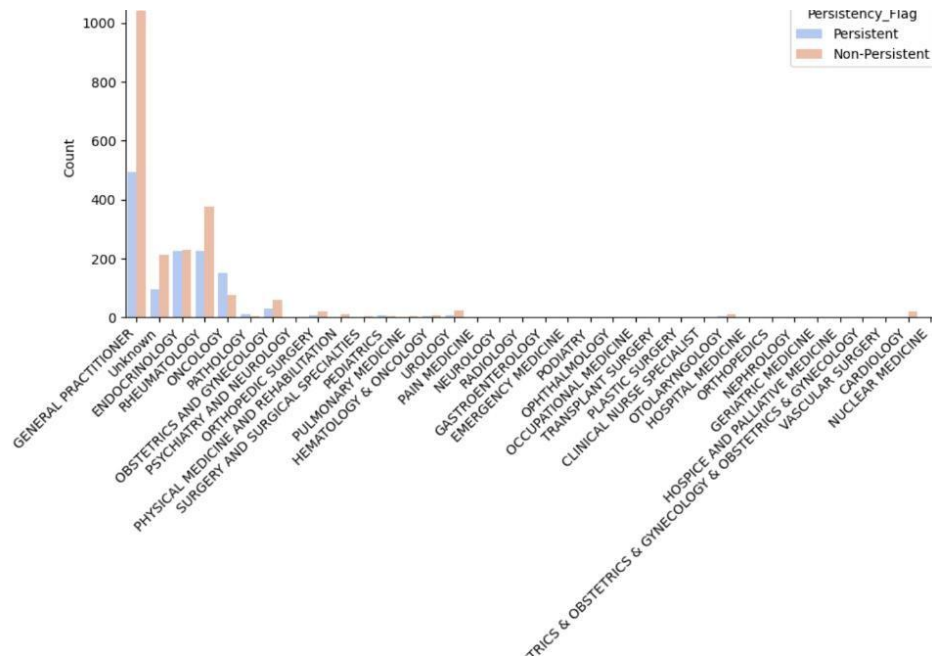
Key Findings:

- **Gender:** Female patients are more represented overall, with slightly higher non-persistence.
- **Race/Ethnicity:** Caucasian and non-Hispanic groups dominate; both have higher non-persistence rates.
- **Region:** Persistency highest in the South, followed by Midwest and West.
- **Age:** Patients aged 65+ remain the most consistent with medication.

Interpretation:

Older age and geographic factors correlate with higher adherence. Younger and more diverse populations may need stronger engagement programs.





Physician & Provider Insights

Findings:

- Highest persistency and non-persistency rates appear among general practitioners and rheumatologists.
- Non-specialist providers record more non-persistent patients.
- Providers classified as OB/GYN/Others/PCP/Unknown treated the largest number of patients overall.

Interpretation:

Specialist physicians generally maintain better patient persistence, likely due to closer follow-up and specific disease focus.

Risk & Adherence Factors

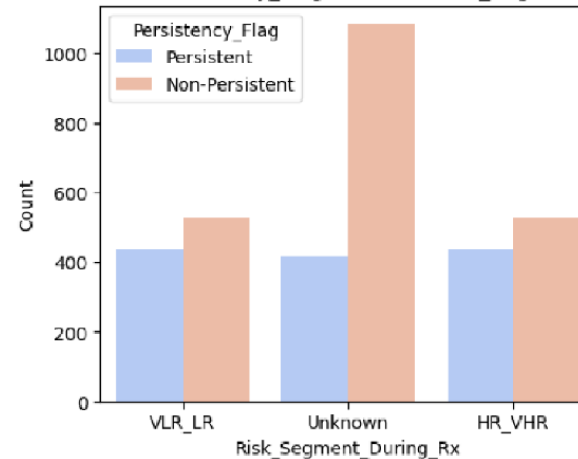
Variables Analyzed:

- *Risk_Segment_Prior_Ntm* – risk level before treatment
- *Risk_Segment_During_Rx* – risk during treatment
- *Change_Risk_Segment* and *Change_T_Score* – improvement indicators
- *Adherent_Flag* – medication adherence

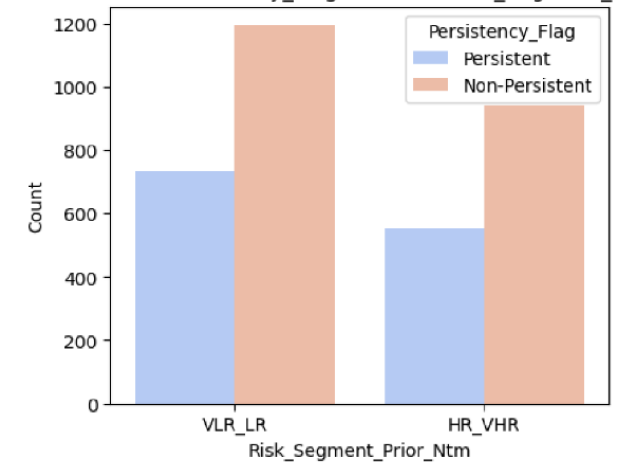
Findings:

- Non-persistence more common among low or unknown risk patients.
- Most patients showed no change in risk or T-score values.
- Even adherent patients sometimes discontinued, showing non-clinical influences.

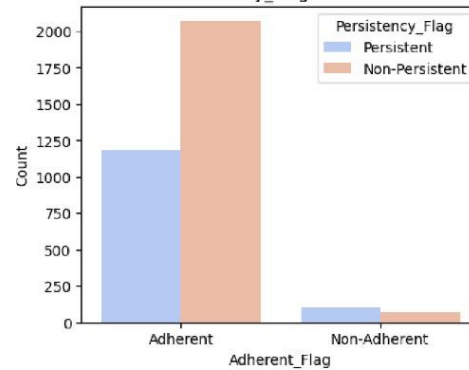
Distribution of Persistency_Flag based on Risk_Segment_During_Rx



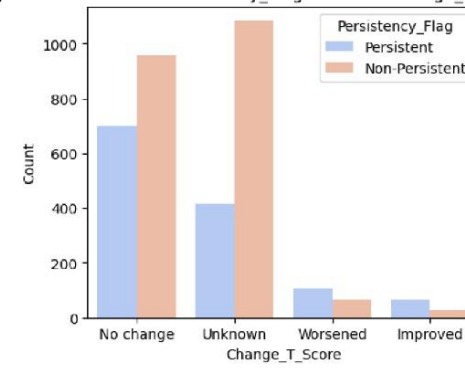
Distribution of Persistency_Flag based on Risk_Segment_Prior_Ntm



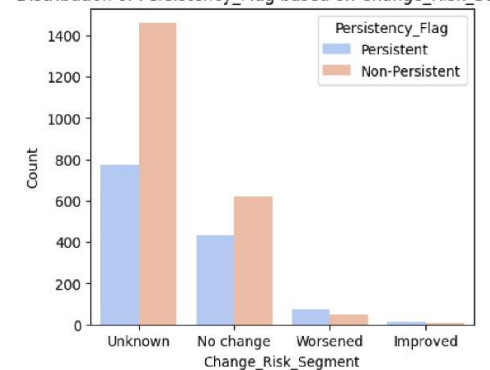
Distribution of Persistency_Flag based on Adherent_Flag



Distribution of Persistency_Flag based on Change_T_Score



Distribution of Persistency_Flag based on Change_Risk_Segment



Model Development

To understand which patients are likely to stay on their prescribed treatment, several models were tested.

What I did:

- Selected the most important patient and physician factors.
- Converted text-based information into numerical form so it can be analyzed.
- Split the data into two parts: one for training the model and one for checking how well it performs.
- Tried several approaches:
 - **Logistic Regression** (simple and clear)
 - **Random Forest** (more complex, tree-based)
 - **Gradient Boosting** (advanced tree-based method)

Goal:

Find the model that gives reliable predictions and is easy for healthcare teams to understand and trust.

Model Performance & Comparison

The three models were compared using accuracy and how well they identify patients who actually stay on treatment.

What this means:

- The two tree-based models (Random Forest & Gradient Boosting) have slightly higher accuracy,
- But they miss many patients who stay on treatment, which is not good for planning or targeting support.
- Logistic Regression identifies these patients much better, giving a healthier balance between all metrics.

Model	Accuracy	Precision	Recall	F1
Logistic Regression (linear)	0.6204	0.4901	0.5866	0.5341
Random Forest (ensemble)	0.6350	0.5138	0.2913	0.3718
Gradient Boosting (boosting)	0.6394	0.5309	0.2362	0.3270

Final Model: Logistic Regression

Why I chose Logistic Regression

- It gives the most balanced and dependable results across all the important indicators.
- It catches more of the patients who actually stay on treatment, which is key for planning interventions.
- It is easy to understand and explain, even for people without a data science background.
- Healthcare projects require transparency - Logistic Regression clearly shows which factors influence the outcome and by how much.

Final status:

The Logistic Regression model is trained, validated, and ready to be used for reporting or integrated into future decision-support tools.

Conclusions



Demographic, physician, and adherence factors are major predictors of drug persistency.



Even high adherence doesn't ensure persistence — social and behavioral drivers remain critical.



Non-specialist physicians see more drop-offs, pointing to the importance of specialized follow-up.



Low-risk patients are more likely to discontinue, suggesting they perceive treatment as less essential.



Takeaway:

Improving persistency requires both clinical monitoring and personalized engagement.

Recommendations

1

1. Patient Engagement:

Offer reminders, counseling, or follow-ups for patients who are at higher risk of stopping treatment.

2

2. Healthcare Provider Insights:

Share key findings with physicians so they better understand which patient groups need extra attention.

3

3. Data Expansion:

Future models could improve by adding more behavioral, medical history, or socioeconomic data.

4

4. Model Improvement:

The model can be refined further by improving the input features and testing slightly stronger methods.

Thank You